Fake News Detection

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Goal

- Rich and unique topic to do machine learning
- We used KAGGLE to find our dataset
- Dataset contains news articles labeled as real and fake along with other variables

Dataset Example

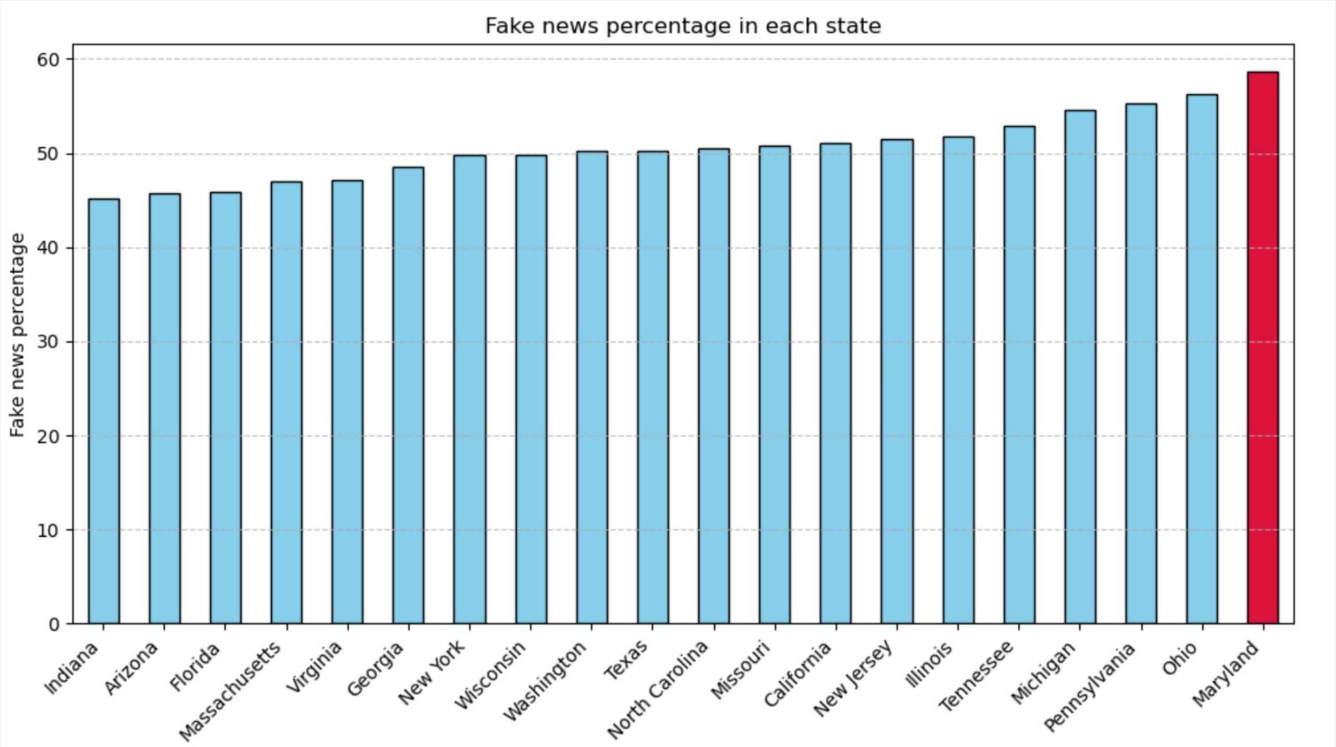
	id	title	author	text	state	date_published	source	category	sentiment_score	word_count	...	num_shares	num_comments	political_bias	fact_
0	1	Breaking News 1	Jane Smith	This is the content of article 1. It contains ...	Tennessee	30-11-2021	The Onion	Entertainment	-0.22	1302	...	47305	450	Center	
1	2	Breaking News 2	Emily Davis	This is the content of article 2. It contains ...	Wisconsin	02-09-2021	The Guardian	Technology	0.92	322	...	39804	530	Left	
2	3	Breaking News 3	John Doe	This is the content of article 3. It contains ...	Missouri	13-04-2021	New York Times	Sports	0.25	228	...	45860	763	Center	

3 rows × 24 columns

Important Variables

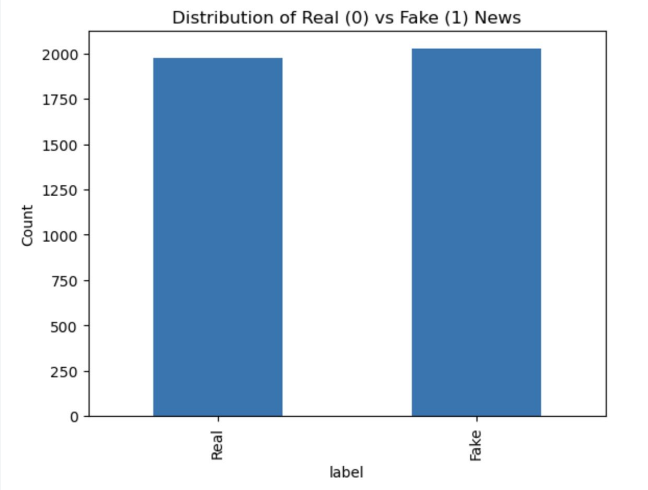
- Sentiment_score
  - Attitude toward article
- Word_count
  - Number of word in article
- Readability_score
  - measures how easy or hard a text is to read
- Num_shares
  - Number of shares of article
- Num_comments
  - Number of comments on the article
- Trust_score
  - Credibility of article
- Source_reputation
  - Credibility based on publisher

Q1. Which state might be populated with the most fake news ?



Q2: Which model performed the best in detecting fake news and how did it work?

Step 1: Check Dataset



Step 2: Feature Selection

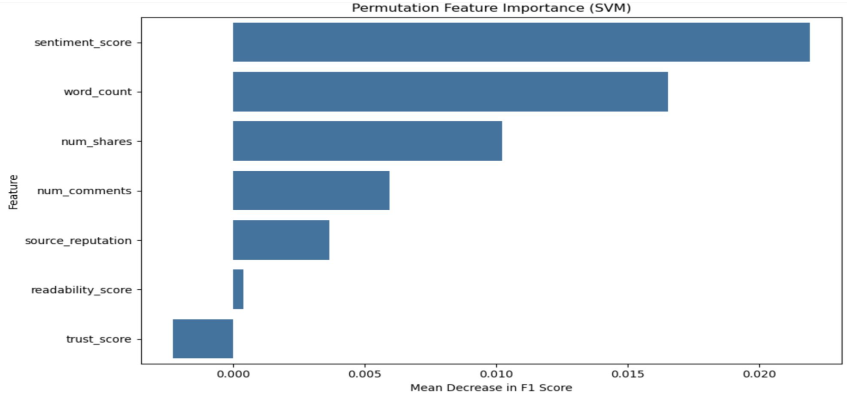
```
features = [
    'sentiment_score',
    'word_count',
    'readability_score',
    'num_shares',
    'num_comments',
    'trust_score',
    'source_reputation']
```

Step 3: Data Split & Compare Models & Feature Permutation

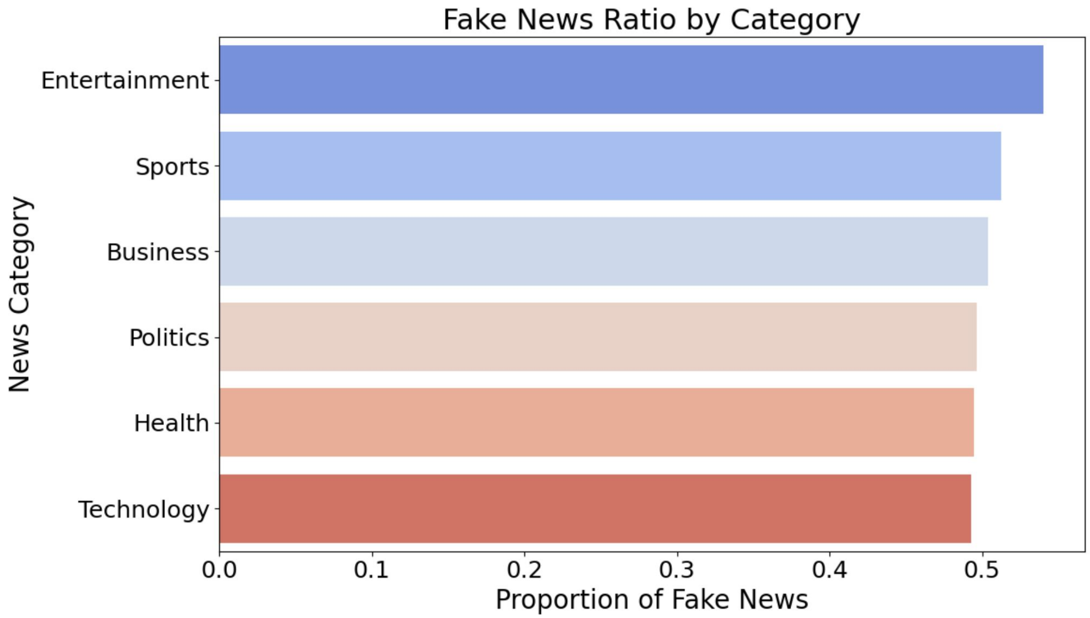
```python
models = {
    'Logistic Regression': {
        'model': LogisticRegression(max_iter=1000),
        'params': {
            'model__C': [0.01, 0.1, 1, 10]
        }
    },
    'Decision Tree': {
        'model': DecisionTreeClassifier(),
        'params': {
            'model__max_depth': [3, 5, 10, None],
            'model__min_samples_split': [2, 5, 10]
        }
    },
    'SVM': {
        'model': SVC(),
        'params': {
            'model__C': [0.1, 1, 10],
            'model__kernel': ['linear', 'rbf']
        }
    },
    'KNN': {
        'model': KNeighborsClassifier(),
        'params': {
            'model__n_neighbors': [3, 5, 7, 9]
        }
    }
}
```

Final: Outcomes from Validation Set

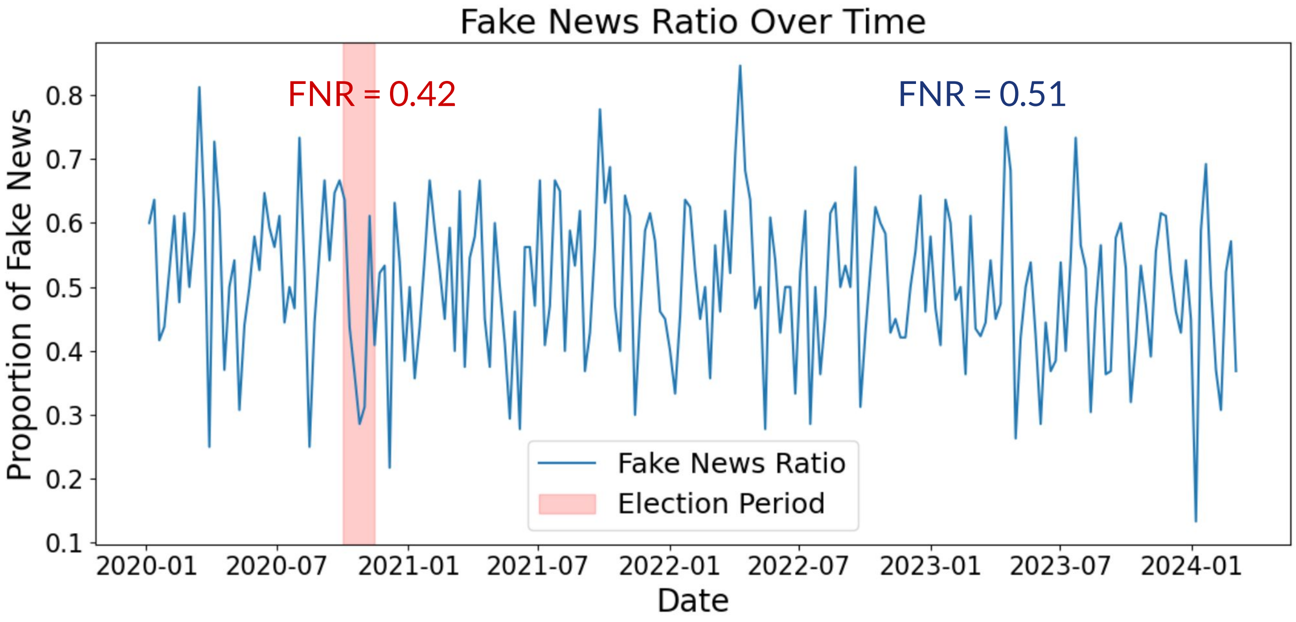
```
Model Performance Summary:

                 Model                                        Best Params  \
2                  SVM         {'model__C': 0.1, 'model__kernel': 'rbf'}
0  Logistic Regression                                 {'model__C': 0.01}
1        Decision Tree  {'model__max_depth': 10, 'model__min_samples_s...
3                  KNN                          {'model__n_neighbors': 9}

   Accuracy  Precision  Recall  F1-Score
2     0.505      0.511   0.922     0.657
0     0.532      0.534   0.718     0.613
1     0.485      0.500   0.568     0.532
3     0.498      0.512   0.515     0.513
```



Q3. Which topics are more likely to be fake or real in the data?



Q4. Does fake news increase during political events?



Conclusion + Next Steps

Key Takeaways

- High recall, but low precision
- Many false positives and negatives

Limitations

- Differences between real/fake news very minimal
- Precision sufferers from labeling real news as fake news